

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 4

SOME CONCEPTS OF ECOLOGY

ADAPTATION

- Each organism is adapted to its particular environment.
- Adaptation is 'the appearance or behaviour or way of life of an organism that allows it to survive in a particular environment.'
- Adaptation is any attribute of the organism that enables the organism to survive and reproduce in its habitat.
- **Examples:**
 - ✓ The Eyelids and Stomach and Hump of a Camel.
 - ✓ When trees grew higher, the Giraffe's neck got longer.
 - ✓ Some desert plants like Opuntia, have no leaves - they are reduced to spines-and the photosynthetic function is taken over by the flattened stems [less leaves = less area available for transpiration].



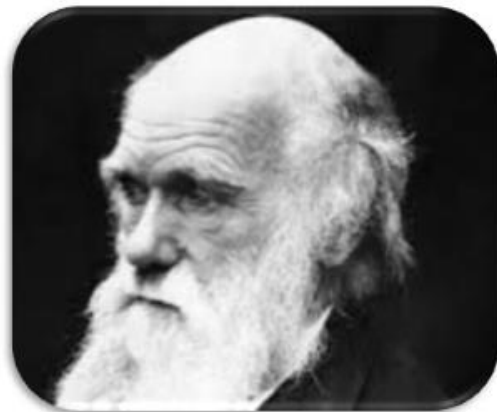
ACCLIMATIZATION

- We need to breathe faster when we are on high mountains. After some days, our body adjusts to the changed conditions on the high mountain.
- Such small changes that take place in the body of an organism over short periods, to overcome small problems due to changes in the surroundings, are called Acclimatization.

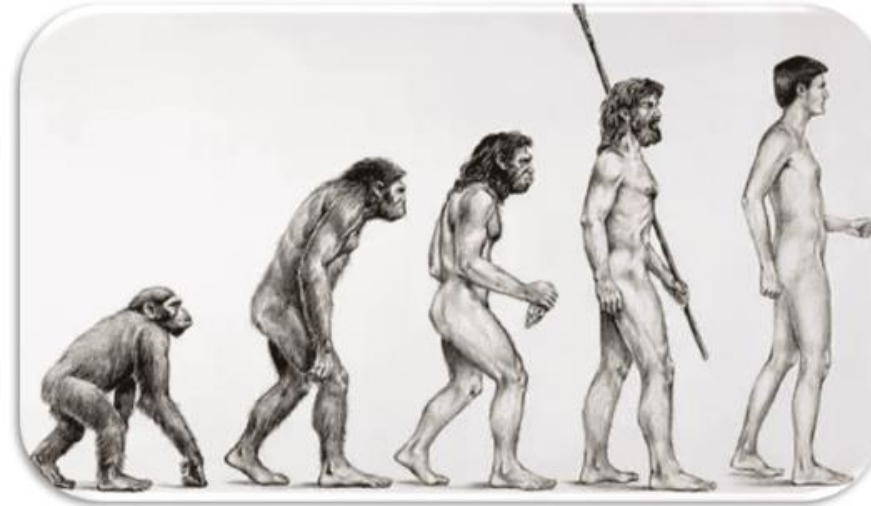


EVOLUTION

- Evolution is the change which gives rise to new species.
- It happens in order to make the organism better suited to the present environment.
- Evolution leads to speciation or formation of new species.

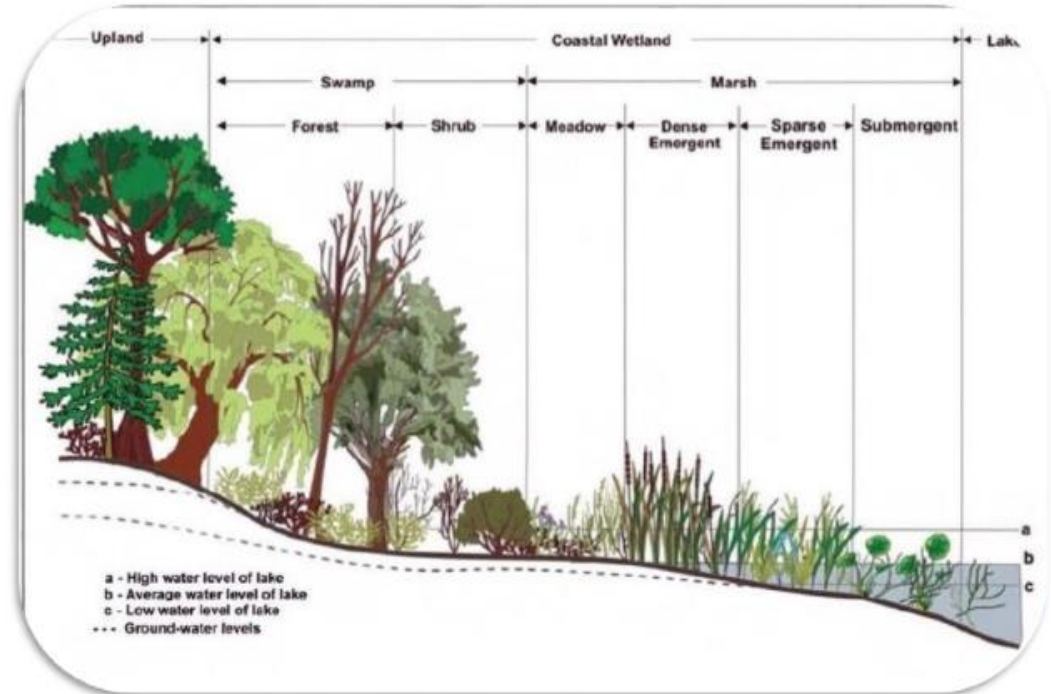


CHARLES DARWIN



ECOTONE

- An Ecotone is a transition area between two ecosystems.
- It is where two communities from two different ecosystems meet and integrate.
- **Examples:**
 - ✓ Wetlands
 - ✓ Mangrove forests represent an Ecotone between marine and terrestrial ecosystem.
 - ✓ Grassland (between forest and desert) and
 - ✓ Estuary (between fresh water and salt water).



ECOTONE – EXAMPLES



MANGROVE FORESTS



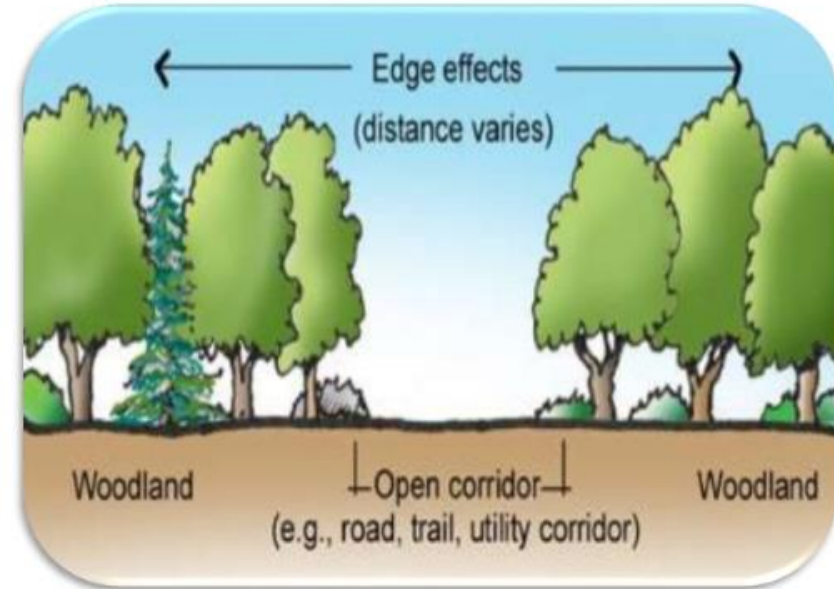
GRASSLAND

CHARACTERISTICS OF ECOTONE

- As it is a zone of transition, it has conditions intermediate to the adjacent ecosystems.
- A well-developed Ecotone contains some organisms which are entirely different from that of the adjoining communities. (Edge Species)

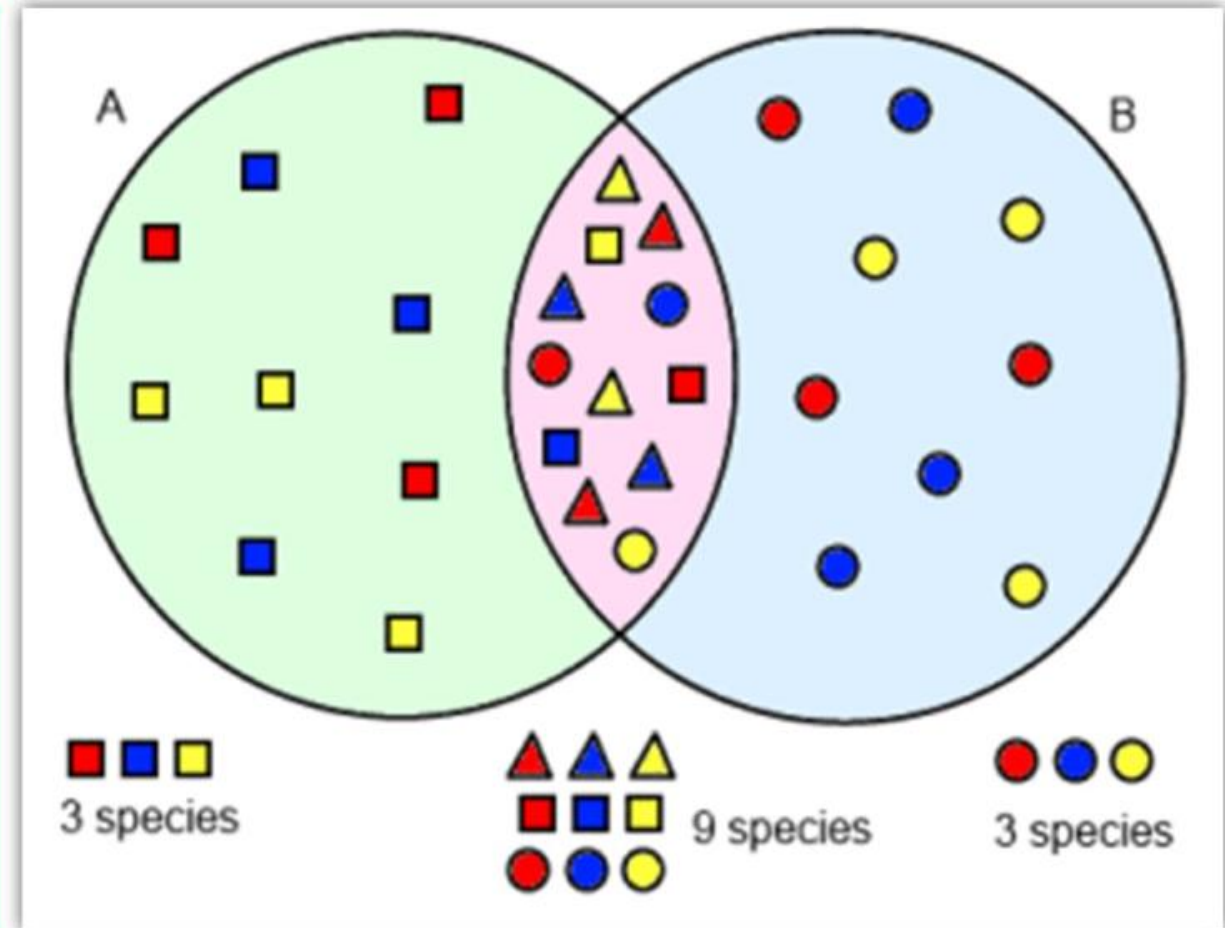
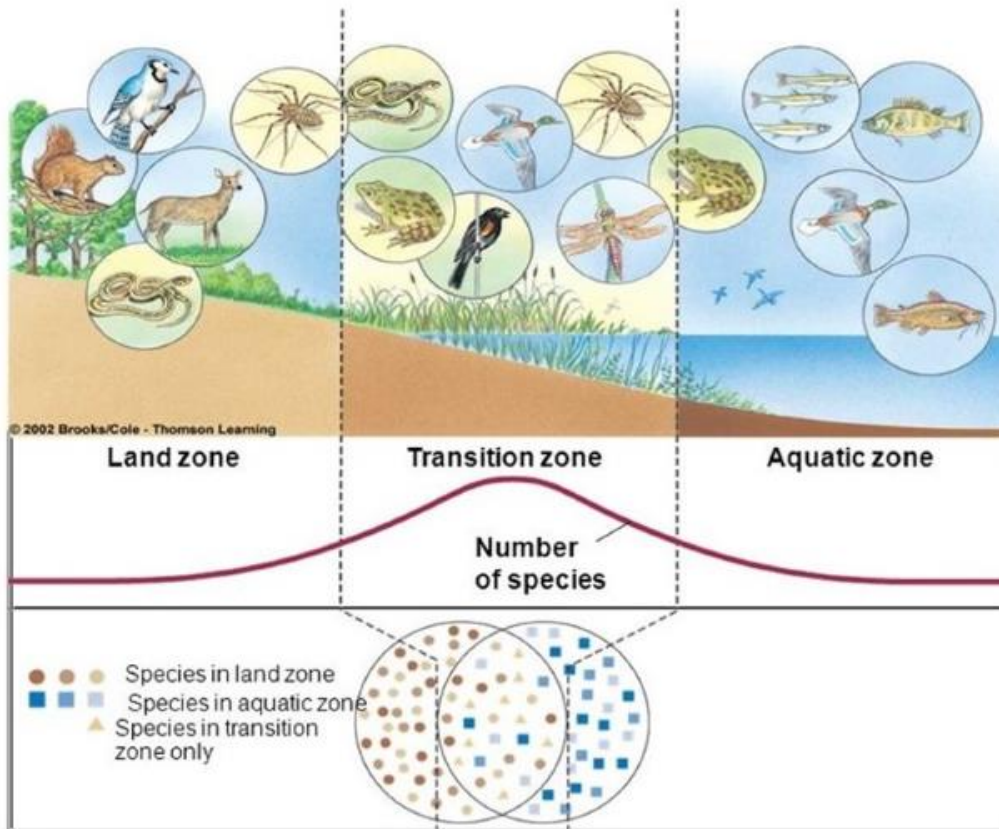
EDGE EFFECT & EDGE SPECIES

- Sometimes the number of species and the population density of some of the species in the Ecotone is much greater than either community. This is called Edge Effect
- The organisms which occur primarily or most abundantly in this zone are known as Edge Species.



CHARACTERISTICS OF ECOTONE

Species Overlap in Ecotones



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